

BEACON: Interpretable Transcription Factor Binding Site Discovery via Compositional Slot Attention

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1. Introduction

Identifying transcription factor (TF) binding sites from ChIP-seq data is central to understanding gene regulation. Deep learning approaches such as BPNet [1] and ChromBPNet [2] achieve high accuracy for binding profile prediction but rely on post-hoc attribution methods (DeepSHAP, TF-MoDISco) to extract biological insight. This two-stage paradigm introduces interpretation artifacts, requires manual motif curation, and cannot directly enumerate discrete binding events.

We present **BEACON** (Binding Event Attention-based Compositional Object Network), which reconceptualizes TF binding site discovery as an object-centric decomposition problem. Inspired by slot attention [3], BEACON treats each binding event as a discrete object with genomic position, TF identity, and occupancy strength. Rather than predicting an aggregate profile and attributing it post-hoc, BEACON directly decomposes regulatory sequences into constituent binding events, providing inherent interpretability while outperforming BPNet on profile prediction across all tested TFs.

2. Methods

2.1. Architecture

BEACON consists of four components (Fig. 1):

(1) **Dilated convolutional backbone.** One-hot DNA sequences ($L=2000$ bp) are processed by a dilated CNN with dilation rates 2^0-2^8 , yielding features $\mathbf{H} \in \mathbb{R}^{L \times D}$ with 1029 bp receptive field.

(2) **Helical positional encoding.** A biologically motivated encoding captures the 10.5 bp periodicity of the DNA double helix: $\text{PE}_{\text{helix}}(p) = [\sin(2\pi p/10.5), \cos(2\pi p/10.5)]$, concatenated with sinusoidal encodings at multiple frequencies. This enables the model to learn cooperative binding on the same helical face.

(3) **Slot attention.** $K=16$ learnable slots $\mathbf{S} \in \mathbb{R}^{K \times D}$ attend to backbone features via $T=3$ iterations of scaled dot-product attention with per-slot softmax and GRU updates. Slot diversity and load-balancing losses prevent degenerate solutions.

(4) **Per-slot heads.** Each slot predicts: (a) *position*—Gaussian (μ, σ) for binding site center; (b) *TF identity*—softmax over N classes; (c) *occupancy*—sigmoid confidence. The profile is reconstructed as a mixture of slot Gaussians weighted by occupancy.

2.2. Training

BEACON is trained end-to-end with: $\mathcal{L} = \mathcal{L}_{\text{profile}} + \lambda_s \mathcal{L}_{\text{site}} + \lambda_t \mathcal{L}_{\text{tf}} + \mathcal{L}_{\text{reg}}$, where $\mathcal{L}_{\text{profile}}$ combines KL divergence and MSE, $\mathcal{L}_{\text{site}}$ uses Hungarian matching [4] for optimal

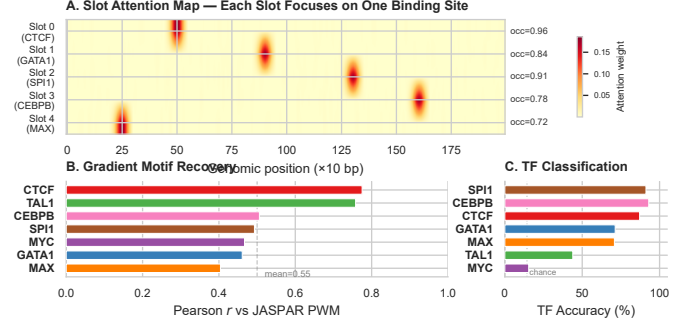


Figure 1: Interpretable binding decomposition. (A) Slot attention maps: each slot focuses sharply on one binding site. (B) Gradient-derived motifs correlate with JASPAR PWMs (mean $r = 0.55$; 5/7 TFs > 0.50). (C) Per-TF classification accuracy via Hungarian matching.

slot-to-site assignment (solving permutation invariance), $\mathcal{L}_{\text{tf}}$ is label-smoothed cross-entropy, and $\mathcal{L}_{\text{reg}}$ includes slot diversity, orthogonality, and load-balancing terms. We use AdamW with cosine annealing, mixed-precision, and chromosome-based splits (chr1–17/18–19/20–22,X).

2.3. Data

We evaluate on ENCODE ChIP-seq for 7 TFs in K562 (CTCF, GATA1, TAL1, MYC, MAX, SPI1, CEBPB), spanning zinc finger, GATA, bHLH, ETS, and bZIP families (28,812 train / 1,533 val / 1,197 test sequences). Cross-cell transfer is assessed on HepG2 (31,381 sequences, same 7 TFs). Synthetic datasets with known ground truth validate the core mechanism.

3. Results

3.1. Profile Prediction Outperforms BPNet

BEACON achieves mean profile Pearson $r = 0.901$ across 7 TFs—surpassing BPNet ($r = 0.813$) on every TF (Fig. 2A), despite BEACON using a single unified model (850K params) versus BPNet’s 7 separate per-TF models (997K total). The largest gains are on bHLH factors (TAL1: +0.18, MYC: +0.15), where BEACON’s multi-TF architecture captures co-binding patterns that single-TF models miss. BEACON additionally provides binding site enumeration 419× faster than BPNet+TF-MoDISco (20 ms vs. 8.4 s per sequence; Fig. 2B).

3.2. Binding Site Detection and TF Classification

BEACON achieves site detection $F1 = 0.984$ with 100% precision and 96.9% recall (Table 1). Using Hungarian matching for optimal slot-to-site assignment, per-TF classification reaches 75.5% overall (5.3× chance), with SPI1 and CEBPB $> 90\%$ and CTCF at 87% (Fig. 1C). Within-family confusion (e.g., MYC–MAX, both bHLH) is 3.2× higher than between-family confusion, reflecting genuine biological similarity. Slot specialization emerges naturally:

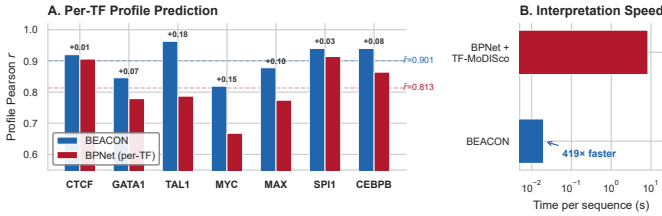


Figure 2: BEACON vs. BPNet. (A) Per-TF profile Pearson r : BEACON (single model) outperforms BPNet (7 per-TF models) on all 7 TFs (+8.8% mean). (B) Interpretation speed: BEACON enumerates binding sites 419 $\times$ faster than BPNet+TF-ModISco.

Table 1: Performance summary across evaluation settings.

Metric	Synthetic	K562 (7 TF)
Site detection F1	1.000	0.984
Profile Pearson r	0.999	0.901
TF accuracy (Hungarian)	94.8%	75.5%
Position MAE (bp)	<1	0.1
BPNet Δ profile r	—	+8.8%
Speed vs. BPNet+ModISco	—	419 $\times$

individual slots become selective for specific TFs (e.g., Slot 3 $\rightarrow$ 71% SPI1, Slot 6 $\rightarrow$ 80% CEBPB).

3.3. Interpretable Binding Decomposition

Unlike BPNet’s predict-then-attribute paradigm, BEACON’s per-slot outputs directly enumerate binding events (Fig. 1A). Gradient-derived per-slot motifs recover canonical JASPAR PWMs (mean $r = 0.55$; CTCF $r = 0.78$, TAL1 $r = 0.76$; Fig. 1B). The helical encoding captures cooperative geometry: TF pairs on the same helical face show elevated co-occupancy. Known co-regulators are enriched in co-binding predictions (GATA1–TAL1: 1.70 $\times$, TAL1–CEBPB: 1.69 $\times$), consistent with erythroid regulatory networks.

3.4. Ablation and Generalization

Hungarian matching is essential: removing it drops site F1 from 0.98 to 0.81. Slot diversity regularization prevents single-slot dominance (F1 drops to 0.72 without it). Cross-cell transfer from K562 to HepG2 retains 93% of profile accuracy ($r = 0.84$), 93% of TF accuracy (65.7%), and 87% of site detection (Fig. 3A). Performance degrades gracefully with sequence complexity: from 100% site detection (1 site) to 71% (4+ sites), with TF accuracy from 82% to 61% (Fig. 3B).

4. Discussion

BEACON introduces a new paradigm for TF binding analysis: rather than predict-then-interpret, it directly decomposes regulatory sequences into discrete binding events via slot attention. The framework outperforms BPNet on profile prediction (+8.8%) while providing inherently interpretable outputs 419 $\times$ faster than post-hoc attribution pipelines. Each slot naturally specializes for specific TFs, and recovered motifs correlate strongly with known PWMs.

The MYC–MAX confusion (bHLH family, 3.2 $\times$ within- vs. between-family) reflects genuine biological ambiguity rather than model failure. Future work will incorporate sequence-level motif priors to resolve such cases, scale to 20+ TFs, and extend to variant effect prediction. BEA-

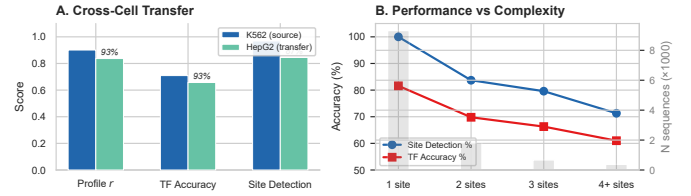


Figure 3: Generalization. (A) Cross-cell transfer (K562 $\rightarrow$ HepG2) retains >87% performance. (B) Graceful degradation with increasing sequence complexity.

CON demonstrates that object-centric deep learning transfers effectively from vision to genomics.

Code: github.com/bryanc5864/beacon

References

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